

Week 3 Module:

Date: 29/06/2026

Topic 1: Merge Sort

Question 1 (Easy) – Rank Students by Marks

A school has recently conducted a scholarship examination for all students of Class 12. The examination department has collected the marks of all students in an array. Due to the large number of participants, manually arranging the marks in ascending order has become difficult.

Your task is to write a program that takes the marks of N students and sorts them in ascending order using the Merge Sort algorithm.

Merge Sort works by repeatedly dividing the array into smaller subarrays until each subarray contains a single element. These smaller subarrays are then merged in a sorted manner to produce the final sorted array.

The school administration wants the marks to be displayed in increasing order so they can easily identify the lowest and highest scoring students and generate performance reports.

Input

- First line contains an integer N, representing the number of students.
- Second line contains N space-separated integers, representing the marks of the students.

Output

- Print the marks in ascending order.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq \text{Marks} \leq 100$

Example

Input:

6
78 45 92 34 67 81

Output:

34 45 67 78 81 92

① ②

Module 3

1) Merge Sort:

What is merge Sort:-

merge Sort is a Sorting algorithm that arranges the elements in ascending or descending order using the divide and Conquer technique.

How merge Sort works.

It follows three Steps.

- 1) Divide: Split the array into two equal halves.
- 2) Conquer: Recursively Sort each half.
- 3) merge: Combine the two Sorted Array halves into one Sorted array.

• Why it is used:-

- Sorts large datasets efficiently.
- It is Stable Sorting algorithm meaning equal elements keep their original order.
- It is useful when Consistent performance is important.

③
Time Complexity →

Best — $O(n \log n)$

Average — $O(n \log n)$

Worst — $O(n \log n)$.

```

1  #include <stdio.h>
2
3  // Function to merge two halves of an array
4  void merge(int arr[], int left, int mid, int right) {
5      int n1 = mid - left + 1;
6      int n2 = right - mid;
7
8      // Temporary arrays
9      int L[n1], R[n2];
10
11     // Copy data to temporary arrays L[] and R[]
12     for (int i = 0; i < n1; i++)
13         L[i] = arr[left + i];
14     for (int j = 0; j < n2; j++)
15         R[j] = arr[mid + 1 + j];
16
17     int i = 0, j = 0, k = left;
18
19     // Merge the temporary arrays back into arr[left..right]
20     while (i < n1 && j < n2) {
21         if (L[i] <= R[j]) {
22             arr[k] = L[i];
23             i++;
24         } else {
25             arr[k] = R[j];
26             j++;
27         }
28         k++;
29     }
30
31     // Copy the remaining elements of L[], if there are any
32     while (i < n1) {
33         arr[k] = L[i];
34         i++;
35         k++;

```

```

28     k++;
29 }
30
31 // Copy the remaining elements of L[], if there are any
32 while (i < n1) {
33     arr[k] = L[i];
34     i++;
35     k++;
36 }
37
38 // Copy the remaining elements of R[], if there are any
39 while (j < n2) {
40     arr[k] = R[j];
41     j++;
42     k++;
43 }
44 }
45
46 // Main function to perform Merge Sort
47 void mergeSort(int arr[], int left, int right) {
48     if (left < right) {
49         // Calculate mid point to avoid overflow
50         int mid = left + (right - left) / 2;
51
52         // Sort first and second halves
53         mergeSort(arr, left, mid);
54         mergeSort(arr, mid + 1, right);
55
56         // Merge the sorted halves
57         merge(arr, left, mid, right);
58     }
59 }
60

```



```

44 }
45
46 // Main function to perform Merge Sort
47 void mergeSort(int arr[], int left, int right) {
48     if (left < right) {
49         // Calculate mid point to avoid overflow
50         int mid = left + (right - left) / 2;
51
52         // Sort first and second halves
53         mergeSort(arr, left, mid);
54         mergeSort(arr, mid + 1, right);
55
56         // Merge the sorted halves
57         merge(arr, left, mid, right);
58     }
59 }
60
61 int main() {
62     int N;
63     scanf("%d", &N); // Number of students
64
65     int marks[N];
66     for(int i = 0; i < N; i++) {
67         scanf("%d", &marks[i]);
68     }
69
70     mergeSort(marks, 0, N - 1);
71
72     for(int i = 0; i < N; i++) {
73         printf("%d ", marks[i]);
74     }
75     printf("\n");
76
77     return 0;
78 }

```

```
56 // Merge the sorted halves
57 merge(arr, left, mid, right);
58 }
59 }
60
61 int main() {
62     int N;
63     scanf("%d", &N); // Number of students
64
65     int marks[N];
66     for(int i = 0; i < N; i++) {
67         scanf("%d", &marks[i]);
68     }
69
70     mergeSort(marks, 0, N - 1);
```

6 78 45 92 34 67 81
34 45 67 78 81 92

...Program finished with exit code 0
Press ENTER to exit console. □

Question 2 (Easy) – Organizing Product Prices

An e-commerce company stores product prices in the order they are added to the platform. Before publishing a seasonal sales report, the company wants all product prices to be arranged in ascending order.

Since the number of products can be very large, an efficient sorting technique is required. Your task is to implement Merge Sort to arrange the prices from the cheapest product to the most expensive product.

The program should divide the list into smaller parts, sort each part, and merge them back together while maintaining the correct order.

Input

- First line contains integer N .
- Second line contains N space-separated product prices.

Output

- Print the sorted prices in ascending order.

Constraints

- $1 \leq N \leq 100000$
- $1 \leq \text{Price} \leq 1000000$

Example

Input:

```
5
2500 1500 4000 3000 1000
```

Output:

```
1000 1500 2500 3000 4000
```



```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  void merge(long long arr[], int left, int mid, int right) {
5      int n1 = mid - left + 1;
6      int n2 = right - mid;
7      long long L[n1], R[n2];
8
9      for (int i = 0; i < n1; i++) L[i] = arr[left + i];
10     for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];
11
12     int i = 0, j = 0, k = left;
13     while (i < n1 && j < n2) {
14         if (L[i] <= R[j]) {
15             arr[k++] = L[i++];
16         } else {
17             arr[k++] = R[j++];
18         }
19     }
20     while (i < n1) arr[k++] = L[i++];
21     while (j < n2) arr[k++] = R[j++];
22 }
23
24 void mergeSort(long long arr[], int left, int right) {
25     if (left < right) {
26         int mid = left + (right - left) / 2;
27         mergeSort(arr, left, mid);
28         mergeSort(arr, mid + 1, right);
29         merge(arr, left, mid, right);
30     }
31 }
32
33 int main() {
34     int N;
35     scanf("%d", &N);

```

```

19     }
20     while (i < n1) arr[k++] = L[i++];
21     while (j < n2) arr[k++] = R[j++];
22 }
23
24 void mergeSort(long long arr[], int left, int right) {
25     if (left < right) {
26         int mid = left + (right - left) / 2;
27         mergeSort(arr, left, mid);
28         mergeSort(arr, mid + 1, right);
29         merge(arr, left, mid, right);
30     }
31 }
32
33 int main() {
34     int N;
35     scanf("%d", &N);
36
37     // Using dynamically allocated array for large constraints
38     long long *prices = (long long *)malloc(N * sizeof(long long));
39
40     for(int i = 0; i < N; i++) {
41         scanf("%lld", &prices[i]);
42     }
43
44     mergeSort(prices, 0, N - 1);
45
46     for(int i = 0; i < N; i++) {
47         printf("%lld ", prices[i]);
48     }
49     printf("\n");
50
51     free(prices);
52     return 0;
53 }

```

```
30     }
31 }
32
33 int main() {
34     int N;
35     scanf("%d", &N);
36
37     // Using dynamically allocated array for large constraints
38     long long *prices = (long long *)malloc(N * sizeof(long long));
39
40     for(int i = 0; i < N; i++) {
41         scanf("%lld", &prices[i]);
42     }
43
44     mergeSort(prices, 0, N - 1);
45 }
```



```
5 2500 1500 4000 3000 1000
1000 1500 2500 3000 4000
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```


Question 3 (Medium) – Sort Employee Records by Salary

A multinational company has employee salary records stored in an array. Before generating annual reports, the Human Resources department wants the salary records arranged in ascending order.

The company expects the number of employees to grow significantly over time, so an efficient sorting algorithm must be used. You are required to implement Merge Sort and display the salaries in increasing order.

Additionally, after sorting, the company wants to know:

1. The lowest salary.
2. The highest salary.
3. The difference between the highest and lowest salary.

Input

- First line contains integer N .
- Second line contains N space-separated salaries.

Output

- First line: Sorted salaries.
- Second line: Lowest salary.
- Third line: Highest salary.
- Fourth line: Salary difference.

Constraints

- $1 \leq N \leq 100000$
- $1000 \leq \text{Salary} \leq 1000000$

Example

Input:

7

45000 25000 70000 50000 30000 90000 60000

Output:

25000 30000 45000 50000 60000 70000 90000

Lowest Salary: 25000

Highest Salary: 90000

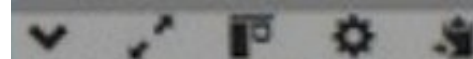
Difference: 65000

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  void merge(int arr[], int left, int mid, int right) {
5      int n1 = mid - left + 1, n2 = right - mid;
6      int L[n1], R[n2];
7
8      for (int i = 0; i < n1; i++) L[i] = arr[left + i];
9      for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];
10
11     int i = 0, j = 0, k = left;
12     while (i < n1 && j < n2) {
13         if (L[i] <= R[j]) arr[k++] = L[i++];
14         else arr[k++] = R[j++];
15     }
16     while (i < n1) arr[k++] = L[i++];
17     while (j < n2) arr[k++] = R[j++];
18 }
19
20 void mergeSort(int arr[], int left, int right) {
21     if (left < right) {
22         int mid = left + (right - left) / 2;
23         mergeSort(arr, left, mid);
24         mergeSort(arr, mid + 1, right);
25         merge(arr, left, mid, right);
26     }
27 }
28
29 int main() {
30     int N;
31     scanf("%d", &N);
32
33     int *salaries = (int *)malloc(N * sizeof(int));
34     for(int i = 0; i < N; i++) {
```

```

27 }
28
29 int main() {
30     int N;
31     scanf("%d", &N);
32
33     int *salaries = (int *)malloc(N * sizeof(int));
34     for(int i = 0; i < N; i++) {
35         scanf("%d", &salaries[i]);
36     }
37
38     mergeSort(salaries, 0, N - 1);
39
40     // 1. Sorted Salaries
41     for(int i = 0; i < N; i++) {
42         printf("%d ", salaries[i]);
43     }
44     printf("\n");
45

```



25000 30000 45000 50000 60000 70000 90000

Lowest Salary: 25000

Highest Salary: 90000

Difference: 65000

...Program finished with exit code 0

Press ENTER to exit console.

Topic 2: Quick Sort

Question 1 (Easy) – Race Timing Analysis

A sports academy records the completion times (in seconds) of runners participating in a race. The coach wants to arrange the timings from the fastest runner to the slowest runner.

To process large datasets efficiently, the academy has decided to use the Quick Sort algorithm. Your task is to sort the race timings in ascending order.

Input

- First line contains integer N .
- Second line contains N space-separated timings.

Output

- Print the timings in ascending order.

Constraints

- $1 \leq N \leq 100000$
- $1 \leq \text{Time} \leq 10000$

Example

Input:

```
6
55 49 62 58 45 51
```

Output:

```
45 49 51 55 58 62
```

(13)

2. Quick Sort:-

• What is Quick Sort:-

Quick Sort is a Sorting algorithm based on the divide and Conquer technique. It work by selecting a ~~pivot~~ Pivot element, placing it in its correct position, and arranging all smaller elements to its left and all ~~ten~~ larger element to its right. The Same process is then repeated for the left and right Subarrays.

How it works.

- Choose a pivot element
- Rearrange the array so that:-
 - element smaller than the pivot are on left
 - element greater than the pivot are ~~on~~ on right
- The ~~two~~ pivot is now in its correct Sorted position.
- ~~Recur Recur~~
- Recursively apply the Same Step to the left and right Subarrays.

(14)

Why it is used:-

- it is very fast for large data sets in practice.
- It sorts in place, so it requires very little extra memory.
- It is widely used in programming libraries and applications where speed is important.

Time Complexity

Best — $O(n \log n)$
average — $O(n \log n)$
worst — $O(n^2)$


```
1 #include <stdio.h>
2
3 // Function to swap two elements
4 void swap(int* a, int* b) {
5     int t = *a;
6     *a = *b;
7     *b = t;
8 }
9
10 // Partition function for Quick Sort
11 int partition(int arr[], int low, int high) {
12     int pivot = arr[high]; // Choosing the last element as pivot
13     int i = (low - 1); // Index of smaller element
14
15     for (int j = low; j <= high - 1; j++) {
16         // If current element is smaller than the pivot
17         if (arr[j] < pivot) {
18             i++;
19             swap(&arr[i], &arr[j]);
20         }
21     }
22     swap(&arr[i + 1], &arr[high]);
23     return (i + 1);
24 }
25
26 // Main Quick Sort function
27 void quickSort(int arr[], int low, int high) {
28     if (low < high) {
29         // Partition index
30         int pi = partition(arr, low, high);
31
32         // Separately sort elements before and after partition
33         quickSort(arr, low, pi - 1);
34         quickSort(arr, pi + 1, high);
35     }
36 }
```

```

21     }
22     swap(&arr[i + 1], &arr[high]);
23     return (i + 1);
24 }
25
26 // Main Quick Sort function
27 void quickSort(int arr[], int low, int high) {
28     if (low < high) {
29         // Partition index
30         int pi = partition(arr, low, high);
31
32         // Separately sort elements before and after partition
33         quickSort(arr, low, pi - 1);
34         quickSort(arr, pi + 1, high);
35     }
36 }
37
38 int main() {
39     int N;
40     scanf("%d", &N);
41
42     int timings[N];
43     for(int i = 0; i < N; i++) {
44         scanf("%d", &timings[i]);
45     }
46
47     quickSort(timings, 0, N - 1);
48
49     for(int i = 0; i < N; i++) {
50         printf("%d ", timings[i]);
51     }
52     printf("\n");
53
54     return 0;
55 }

```

```

26 // Quick Sort function
27 void quickSort(int arr[], int low, int high) {
28     if (low < high) {
29         // Partition index
30         int pi = partition(arr, low, high);
31
32         // Separately sort elements before and after partition
33         quickSort(arr, low, pi - 1);
34         quickSort(arr, pi + 1, high);
35     }
36 }
37
38 int main() {
39     int N;
40     scanf("%d", &N);
41
42     int timings[N];
43     for(int i = 0; i < N; i++) {
44         scanf("%d", &timings[i]);
45     }
46
47     quickSort(timings, 0, N - 1);

```



6 55 49 62 58 45 51

15 49 51 55 58 62

...Program finished with exit code 0

Press ENTER to exit console.

Question 2 (Easy) – Organizing Library Book IDs

A library stores books using unique identification numbers. Due to years of additions and removals, the IDs are no longer arranged properly.

The librarian wants all IDs sorted in ascending order before migrating the data to a new management system. Implement Quick Sort to arrange the IDs efficiently.

Input

- First line contains integer N .
- Second line contains N space-separated book IDs.

Output

- Print the sorted IDs.

Constraints

- $1 \leq N \leq 100000$
- $1 \leq ID \leq 10^9$

Example

Input:

5
1025 1001 1050 1010 1005

Output:

1001 1005 1010 1025 1050

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 void swap(long long* a, long long* b) {
5     long long t = *a;
6     *a = *b;
7     *b = t;
8 }
9
10 int partition(long long arr[], int low, int high) {
11     long long pivot = arr[high];
12     int i = (low - 1);
13
14     for (int j = low; j <= high - 1; j++) {
15         if (arr[j] < pivot) {
16             i++;
17             swap(&arr[i], &arr[j]);
18         }
19     }
20     swap(&arr[i + 1], &arr[high]);
21     return (i + 1);
22 }
23
24 void quickSort(long long arr[], int low, int high) {
25     if (low < high) {
26         int pi = partition(arr, low, high);
27         quickSort(arr, low, pi - 1);
28         quickSort(arr, pi + 1, high);
29     }
30 }
31
32 int main() {
33     int N;
34     scanf("%d", &N);
35
36     // Quick Sort can be used on 1D array using long long for 2-64 bit
```

```

17     swap(&arr[i], &arr[j]);
18 }
19 }
20 swap(&arr[i + 1], &arr[high]);
21 return (i + 1);
22 }
23
24 void quickSort(long long arr[], int low, int high) {
25     if (low < high) {
26         int pi = partition(arr, low, high);
27         quickSort(arr, low, pi - 1);
28         quickSort(arr, pi + 1, high);
29     }
30 }
31
32 int main() {
33     int N;
34     scanf("%d", &N);
35
36     // Book IDs can be up to 10^9, using long long for safety
37     long long *bookIDs = (long long *)malloc(N * sizeof(long long));
38     for(int i = 0; i < N; i++) {
39         scanf("%lld", &bookIDs[i]);
40     }
41
42     quickSort(bookIDs, 0, N - 1);
43
44     for(int i = 0; i < N; i++) {
45         printf("%lld ", bookIDs[i]);
46     }
47     printf("\n");
48
49     free(bookIDs);
50     return 0;
51 }

```



```

24 void quickSort(long long arr[], int low, int high) {
25     if (low < high) {
26         int pi = partition(arr, low, high);
27         quickSort(arr, low, pi - 1);
28         quickSort(arr, pi + 1, high);
29     }
30 }
31
32 int main() {
33     int N;
34     scanf("%d", &N);
35
36     // Book IDs can be up to 10^9, using long long for safety
37     long long *bookIDs = (long long *)malloc(N * sizeof(long long));
38     for(int i = 0; i < N; i++) {
39         scanf("%lld", &bookIDs[i]);
40     }
41
42     quickSort(bookIDs, 0, N - 1);
43

```


input

```

5 1025 1001 1050 1010 1005
1001 1005 1010 1025 1050

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```

Question 3 (Medium) – Online Store Sales Ranking

An online marketplace tracks the daily sales made by different sellers. At the end of each week, management wants to identify the best-performing sellers.

You are given the total sales made by N sellers. Using Quick Sort, arrange the sales figures in descending order.

After sorting, display:

1. Sorted sales figures.
2. Top 3 highest sales values.
3. Average of the top 3 sales values.

Input

- First line contains integer N .
- Second line contains N space-separated sales values.

Output

- Sorted sales figures in descending order.
- Top 3 sales values.
- Average of top 3 sales.

Constraints

- $3 \leq N \leq 100000$
- $1 \leq \text{Sales} \leq 1000000$

Example

Input:

6
1200 5000 3000 2500 7000 4500

Output:

7000 5000 4500 3000 2500 1200
Top 3: 7000 5000 4500
Average: 5500

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 void swap(long long* a, long long* b) {
5     long long t = *a;
6     *a = *b;
7     *b = t;
8 }
9
10 // Modified Partition function for Descending Order
11 int partitionDescending(long long arr[], int low, int high) {
12     long long pivot = arr[high];
13     int i = (low - 1);
14
15     for (int j = low; j <= high - 1; j++) {
16         // Change logic to greater than for descending
17         if (arr[j] > pivot) {
18             i++;
19             swap(&arr[i], &arr[j]);
20         }
21     }
22     swap(&arr[i + 1], &arr[high]);
23     return (i + 1);
24 }
25
26 void quickSortDescending(long long arr[], int low, int high) {
27     if (low < high) {
28         int pi = partitionDescending(arr, low, high);
29         quickSortDescending(arr, low, pi - 1);
30         quickSortDescending(arr, pi + 1, high);
31     }
32 }
33
34 int main() {
35     int N;
36     scanf("%d", &N);

```



```
34 int main() {
35     int N;
36     scanf("%d", &N);
37
38     long long *sales = (long long *)malloc(N * sizeof(long long));
39     for(int i = 0; i < N; i++) {
40         scanf("%lld", &sales[i]);
41     }
42
43     // Sort in Descending order
44     quickSortDescending(sales, 0, N - 1);
45
46     // 1. Print sorted sales
47     for(int i = 0; i < N; i++) {
48         printf("%lld ", sales[i]);
49     }
50     printf("\n");
51
52     // 2. Print Top 3
53     printf("Top 3: ");
54     for(int i = 0; i < 3 && i < N; i++) {
55         printf("%lld ", sales[i]);
56     }
57     printf("\n");
58
59     // 3. Print Average of top 3
60     long long sum = 0;
61     int count = (N >= 3) ? 3 : N;
62     for(int i = 0; i < count; i++) {
63         sum += sales[i];
64     }
65     printf("Average: %lld\n", sum / count);
66
67     free(sales);
68     return 0;
69 }
```

```

41     }
42
43     // Sort in Descending order
44     quickSortDescending(sales, 0, N - 1);
45
46     // 1. Print sorted sales
47     for(int i = 0; i < N; i++) {
48         printf("%lld ", sales[i]);
49     }
50     printf("\n");
51
52     // 2. Print Top 3
53     printf("Top 3: ");
54     for(int i = 0; i < 3 && i < N; i++) {
55         printf("%lld ", sales[i]);
56     }
57     printf("\n");
58
59     // 3. Print Average of top 3
60     long long sum = 0;
61     int count = (N >= 3) ? 3 : N;

```



```

6 1200 5000 3000 2500 7000 4500
7000 5000 4500 3000 2500 1200
Top 3: 7000 5000 4500
Average: 5500

```